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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/071,733	03/05/2025	Xin WU	QFT25324PCTCON	6793

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EXAMINER	
LEE, DOUGLAS	

ART UNIT	PAPER NUMBER
1714	

NOTIFICATION DATE	DELIVERY MODE
10/29/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 19/071,733	Applicant(s) WU et al.	
	Examiner DOUGLAS LEE	Art Unit 1714	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 3/5/25.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 3/5/25 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 112

2. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

3. Claims 17 and 20 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

4. Claim 17 recites “the first condition,” “the second condition,” “the third condition” and “the fourth condition.” There is insufficient antecedent basis for these limitations in the claim because claim 17 depends upon claim 15 and not claim 16 where the recitations are made. For purposes of examination, claim 17 will be interpreted as depending on claim 16. Appropriate correction is required.

5. Claim 20 recites “the first condition,” “the second condition,” “the third condition” and “the fourth condition.” There is insufficient antecedent basis for these limitations in the claim because claim 20 depends upon claim 18 and not claim 19 where the recitations are made. For purposes of examination, claim 20 will be interpreted as depending on claim 19. Appropriate correction is required.

Claim Rejections - 35 USC § 102

6. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

7. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

8. Claim(s) 1-3, 6-8, 12, 13 and 15-20 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipat4ed by U.S. Patent App. Pub. No. 2020/0077863 to Huang et al.

9. As to claims 1, 15 and 18, Huang discloses a control method for a cleaning robot comprising: receiving a weekly cleaning constraining condition input by a user, wherein the weekly cleaning constraining condition represents a condition to be satisfied when the cleaning robot cleans, in a pre-set period, one or more sub-zones of a plurality of sub-zones corresponding to a work zone included in a map of the work zone, wherein the pre-set period is an integer multiple of a week (see Huang paragraphs [0043], [0046]-[0049] disclosing receiving various weekly cleaning constraining condition inputs by the user within a pre-set period of one or more sub-zones of a plurality of sub-zones wherein the pre-set period can be an integer multiple of a week); determining weekly plan options that satisfy the weekly cleaning constraining condition and providing the weekly plan options to the user (see Huang paragraphs [0043]-[0049] where various cleaning schedules are created, compared and presented to the user for possible selection); obtaining a to-be-executed weekly plan option selected by the user from the

weekly plan options (see, e.g., Huang paragraph [0046] where the user can select the desired cleaning election); and controlling the cleaning robot to execute a cleaning task in the pre-set period according to the to-be-executed weekly plan option (see Huang paragraph [0044] where the robot executes the cleaning mission according to the schedule). Regarding claim 15, Huang discloses that the control method can be performed by a control apparatus for a cleaning robot (see Huang paragraphs [0057]-[0059] disclosing one or more programmable processors executing one or more computer programs to perform the functions described of the cleaning robot). Regarding claim 18, Huang discloses a non-transitory computer-readable storage medium storing instructions, which when executed by a processor, cause the processor to perform a control method for a cleaning robot, the control method comprising the control method of claim 1 above (see Huang paragraphs [0057]-[0059]).

10. As to claim 2, Huang discloses that the weekly cleaning constraining condition can include a first condition and/or a second condition wherein the first condition relates to a number of cleanings to be performed in the pre-set period and the second condition relates to an area to be cleaned in the pre-set period (see Huang paragraph [0048], [0053] disclosing number of cleaning passes can be set and paragraph [0050] where individual rooms can be selected).

11. As to claim 3, Huang discloses that the weekly constraining condition can include a third condition wherein the third condition relates to a cleaning time in which the cleaning is to be performed in the pre-set period (see Huang paragraphs [0047]-[0049] where the day and time of the week can be set).

12. As to claim 5, Huang discloses that the weekly cleaning constraining condition can include a fifth condition, wherein the fifth condition relates to a moving mode in the pre-set period (see Huang paragraph [0043] where cleaning parameters can include spot cleaning and edge cleaning).

13. As to claim 6, Huang discloses that the plurality of sub-zones can be divided into a plurality of categories and wherein the first condition can be configured to specify a total number of cleanings to be

performed in the pre-set period for a corresponding category of sub-zones and/or to specify a number of cleanings to be performed on each day of the pre-set period for a corresponding category of sub-zones (see Huang Fig. 8 and paragraph [0053] where the number of cleanings can be based on the size of the rooms or in places where deep cleaning is needed).

14. As to claim 7, Huang discloses that the second condition can be configured to specify a first area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed in the pre-set period and/or a second area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed on a single day, wherein the first area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones and a number of cleanings to be performed in the pre-set period corresponding to each category of sub-zones and wherein the second area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones in which cleaning is to be performed on a single day and a number of cleanings to be performed on a single day for each corresponding category of sub-zones (see Huang paragraphs [0050] and [0053]-[0056] disclosing that the cleaning schedule (read as second condition) can be configured to specify specific rooms and the number of cleanings for each room (read as equal to a sum of products where each product is a product of an area of each category (i.e., each room) and a number of cleanings to be performed in each room).

15. As to claim 8, Huang discloses that the third condition can be configured to specify a cleaning time of a pre-set day of the pre-set period being the shortest or a number of interval days between cleanings for a corresponding category or subzones (see Huang paragraphs [0047]-[0049] where a cleaning time of a pre-set day can be selected or specific days of the week can be selected).

16. As to claim 10, Huang discloses that the moving mode can include at least one of an edge following mode (see Huang paragraph [0043] disclosing that the cleaning parameter can include edge

cleaning). It is noted that the return-to-recharge mode is not a required moving mode since the claim recites “at least one of…”.

17. As to claim 12, Huang discloses that the weekly plan options can change as the weekly cleaning constraining condition changes, and the control method further comprises in response to a change in the to-be-executed weekly plan option, controlling the cleaning robot to execute new cleaning task according to to-be-executed weekly plan option in the pre-set period (see Huang paragraphs [0046]-[0054] where the user can add and customize additional cleaning schedules).

18. As to claim 13, Huang discloses that when there is a conflict between at least two conditions included in the weekly cleaning constraining condition, performing an error processing corresponding to the conflict (see Huang paragraph [0043]).

19. As to claims 16 and 19, Huang discloses that the weekly cleaning constraining condition can include one or more of the following conditions: a first condition and/or a second condition wherein the first condition relates to a number of cleanings to be performed in the pre-set period and the second condition relates to an area to be cleaned in the pre-set period (see Huang paragraph [0048], [0053] disclosing number of cleaning passes can be set and paragraph [0050] where individual rooms can be selected); and/or a third condition wherein the third condition relates to a cleaning time in which the cleaning is to be performed in the pre-set period (see Huang paragraphs [0047]-[0049] where the day and time of the week can be set).

20. As to claims 17 and 20, Huang discloses that the plurality of sub-zones can be divided into a plurality of categories and wherein the first condition can be configured to specify a total number of cleanings to be performed in the pre-set period for a corresponding category of sub-zones and/or to specify a number of cleanings to be performed on each day of the pre-set period for a corresponding category of sub-zones (see Huang Fig. 8 and paragraph [0053] where the number of cleanings can be based on the size of the rooms or in places where deep cleaning is needed). Huang also discloses that

the second condition can be configured to specify a first area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed in the pre-set period and/or a second area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed on a single day, wherein the first area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones and a number of cleanings to be performed in the pre-set period corresponding to each category of sub-zones and wherein the second area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones in which cleaning is to be performed on a single day and a number of cleanings to be performed on a single day for each corresponding category of sub-zones (see Huang paragraphs [0050] and [0053]-[0056] disclosing that the cleaning schedule (read as second condition) can be configured to specify specific rooms and the number of cleanings for each room (read as equal to a sum of products where each product is a product of an area of each category (i.e., each room) and a number of cleanings to be performed in each room). Huang further discloses that the third condition can be configured to specify a cleaning time of a pre-set day of the pre-set period being the shortest or a number of interval days between cleanings for a corresponding category or subzones (see Huang paragraphs [0047]-[0049] where a cleaning time of a pre-set day can be selected or specific days of the week can be selected). It is noted that claims 16 and 19 disclose "one or more of the following conditions" and does not require all of the conditions.

Claim Rejections - 35 USC § 103

21. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

22. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

23. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

24. This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

25. Claim(s) 4, 9, 11 and 14 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Patent App. Pub. No. 2020/0077863 to Huang et al. as applied to claim 1 above, and further in view of U.S. Patent App. Pub. No. 2018/0074508 to Kleiner et al.

26. Huang is relied upon as discussed above with respect to the rejection of claim 1.

27. As to claims 4 and 9, Huang does not explicitly disclose that the weekly cleaning constraining condition also includes a fourth condition and wherein the fourth condition relates to a cleaning order in which cleanings are to be performed in the pre-set period and the fourth condition is configured to specify an order in which a plurality of sub-zones of a same category are to be cleaned and/or an order in which sub-zones of a plurality of categories are to be cleaned on the same day. Kleiner discloses a similar method wherein a specific order of subregions are to be cleaned on the same day (see Kleiner paragraphs [0151], [0162], [0195]). It would have been obvious to one of ordinary skill in the art at the time of filing to include a fourth condition relating to the cleaning order in which the cleanings are to be performed and on which days as disclosed by Kleiner as is known in the art and to optimize the cleaning schedule of the robot.

28. As to claim 11, since Haung discloses that the robot controlled such that the cleaning task is executed at the scheduled pre-set time (see Huang paragraph [0044]), it is reasonably expected that the robot would be directly controlled to execute the cleaning task as scheduled. Furthermore, Kleiner discloses that it is known in the art to track the cleaning path of the robot such that the robot does not traverse a previously cleaned area (see Kleiner paragraph [0229]) and it would have been obvious to one of ordinary skill in the art at the time of filing to determine whether an area has already been cleaned as disclosed by Kleiner in order to optimize the cleaning path of the robot.

29. As to claim 14, Huang does not explicitly disclose based on pre-set or user-set priorities of conditions included in the weekly cleaning constraining condition, determining the weekly plan options on an order from high priority to low priority. Kliner discloses a similar method wherein the user can set the order from high priority to low priority when determining the weekly plan options (see Kleiner paragraphs [0151], [0162], [0195]). It would have been obvious to one of ordinary skill in the art at the time of filing to include user-set priorities in determining the weekly plan options as disclosed by Kleiner in order to allow the user to optimize the cleaning as disclosed by Kleiner.

Conclusion

30. Any inquiry concerning this communication or earlier communications from the examiner should be directed to DOUGLAS LEE whose telephone number is (571)270-3296. The examiner can normally be reached M-F 7:30-4:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kaj Olsen can be reached at 571-272-1344. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/DOUGLAS LEE/
Primary Examiner, Art Unit 1714